CREDIT CARD FRAUD DETECTION SYSTEM

A Tri-Model Architecture Leveraging Decision Trees, XGBoost, and Heterogeneous Graph Neural Networks for Transaction Security

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Introduction & Problem Statement

- **The Problem:** Credit card fraud costs the global economy billions annually.
- **The Challenge:** Fraudulent patterns are "needles in a haystack" (imbalanced data) and are constantly evolving.
- **The Goal:** Build a system that not only looks at transaction numbers but also at how users, devices, and cards are interconnected.

Dataset Overview

- **Source:** IEEE Computational Intelligence Society (IEEE-CIS) dataset from Kaggle.
- **Complexity:** Contains two main tables: Transactions (amounts, timestamps) and Identity (device info, network details).
- **Scale:** Over 500,000 transactions with hundreds of features (V-features, C-features, D-features about 870 columns).
- **Imbalance:** Highlighting that only a tiny percentage of the data is actually fraud.

EDA

- Dataset Overview:
  Total Transactions: 590,540 (Original) | Notebook Sample: 177,162 (30%)
  Dimensionality: 434 Initial Features (Transaction + Identity data join).
- Target Class Imbalance - Legitimate: 96.42% | Fraudulent: 3.58%
- Imbalance Ratio: 1 : 26.9 (Requires stratified sampling).
- Missing Value Analysis: Identified 214 columns with >50% missing data (primarily in Vesta and Identity features).
- High sparsity in "V" features suggests the need for dimensionality reduction (PCA).
- Statistical Findings: Transaction amounts follow a long-tail distribution (Outliers identified at $1,104+).
- Temporal analysis reveals cyclical "midnight" spikes in fraudulent attempts.

Target Class Distribution

Feature Engineering & Preprocessing

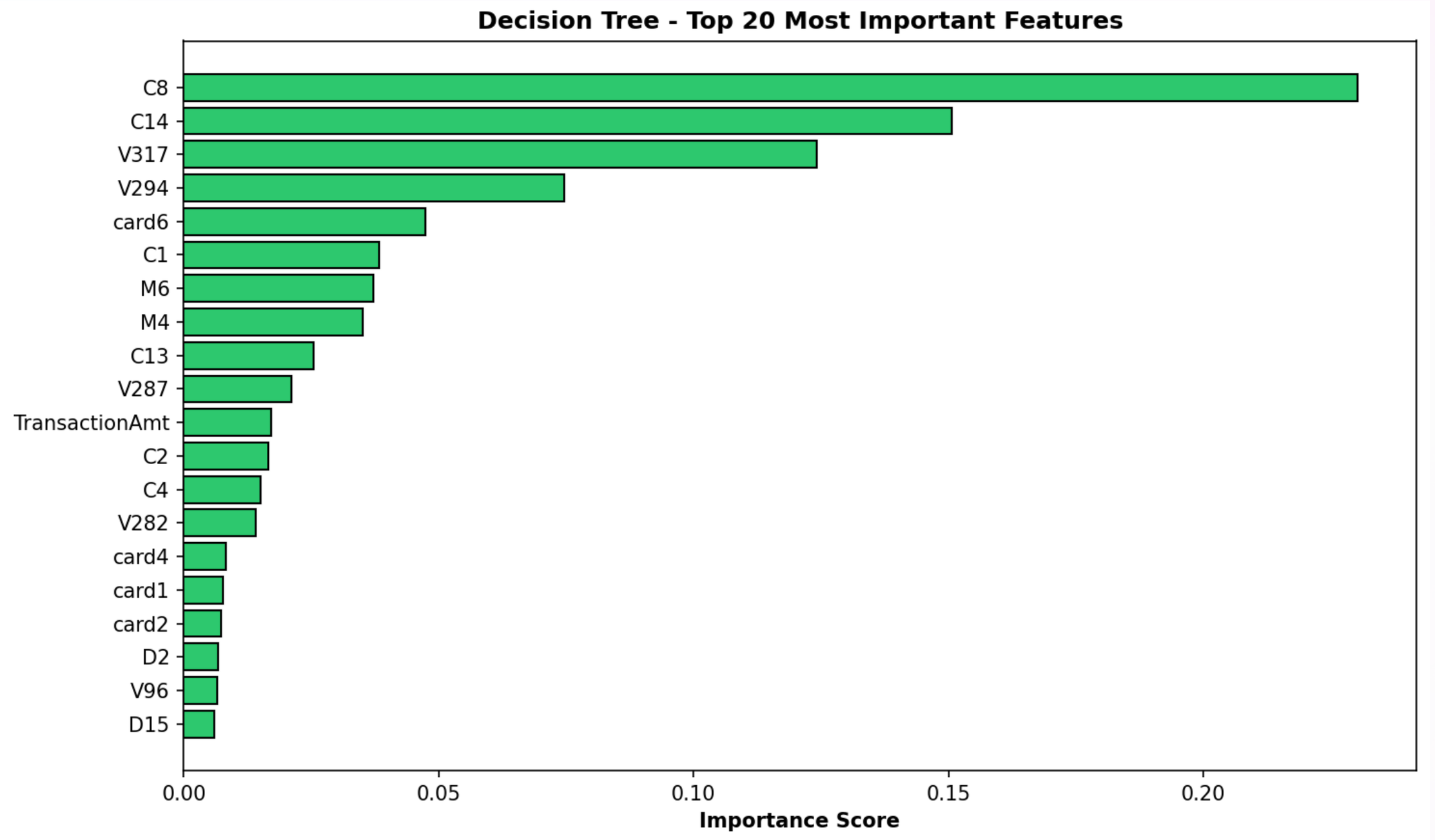
1. Data Cleaning:
   Memory Optimization: Downcasted numeric types, reducing RAM usage by ~60%.
   Missingness Strategy: Median imputation for numeric gaps; "Unknown" tagging for categorical text.
   Outlier Capping: Applied 99th percentile clipping ($1,104) to stabilize model gradients.

2. Feature Transformation & Scaling:
   Z-Score Scaling: Normalized all features to Mean=0, Variance=1.
   Label Encoding: Transformed 9 categorical features (DeviceType, card4, etc.) into numeric inputs.

3. Dimensionality Reduction (PCA):
   Condensed 339 "V-features" into 50 Principal Components (PCA applied only for XGBoost model)
   Maintains 90%+ variance while removing noise and multi-collinearity.

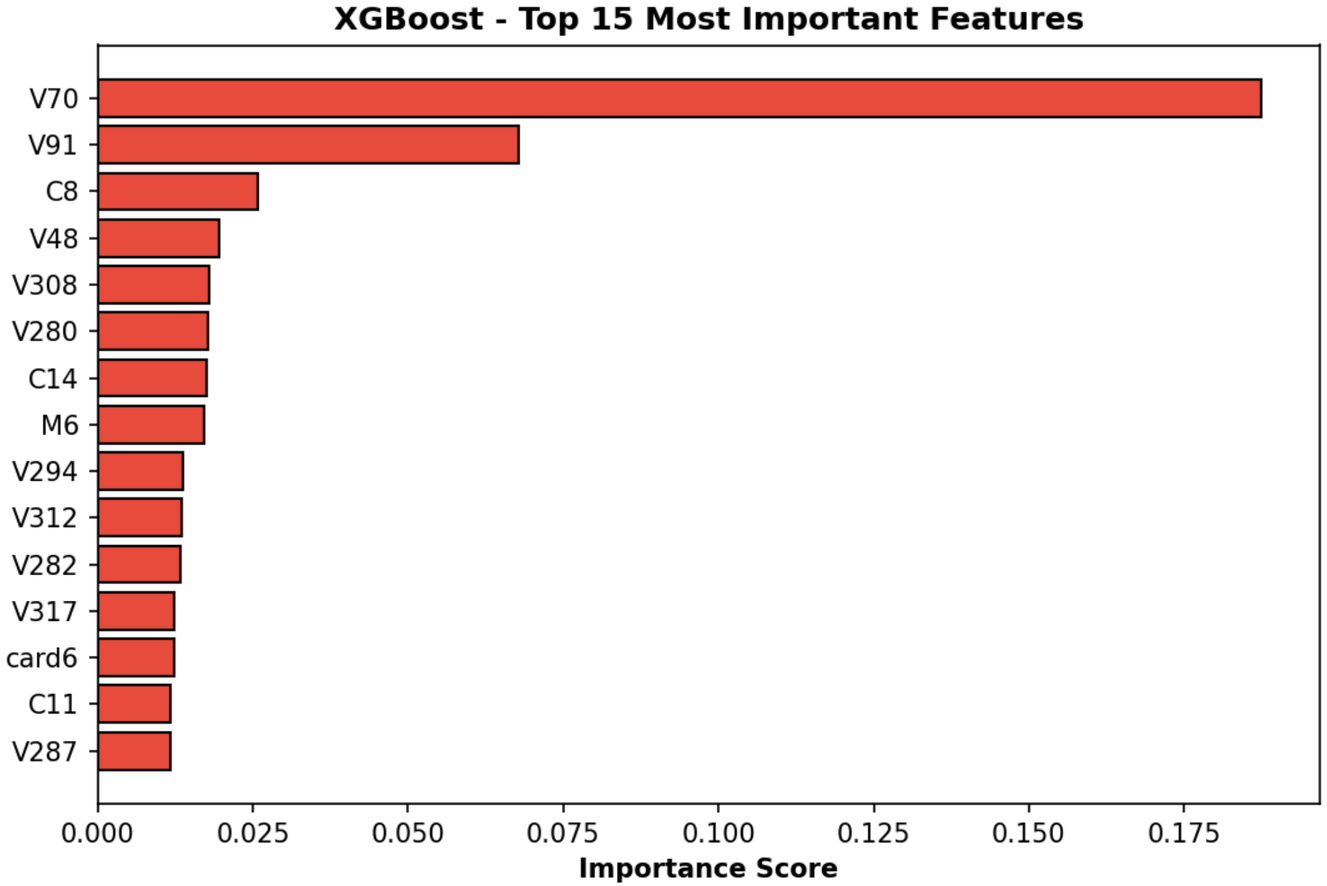
Decision Tree (Baseline Model)

- Model Objectives:
  Establish a "glass-box" baseline to understand feature importance.
  Implement SMOTE (Synthetic Minority Over-sampling Technique) to address the 1:27 class imbalance.
- Training Configuration:
  Algorithm: Decision Tree Classifier (max_depth=12).
  Resampling: Increased fraud representation from 4,439 to 59,786 samples using synthetic generation.
  Weighting: Utilized class_weight='balanced' to further penalize misclassified fraud.
- Baseline Results (Test Set):
  Fraud Detection Rate (Recall): 54.57% (519/951 frauds detected).
  Precision: 29.29% (Low precision due to aggressive flagging).
  ROC-AUC: 0.8390.
- The model successfully identified over half of the fraud cases but produced 1,253 false alarms.
- Top Predictors: Features related to Transaction Amount and Card Identity proved most influential.
- Conclusion: Serves as a benchmark; more advanced models (XGBoost/HGNN) are required to reduce "Customer Friction" (False Positives).


Decision Tree - Top 20 Most Important Features
C8
C14
V317
V294
card6
C1
M6
M4
C13
V287
TransactionAmt
C2
C4
V282
card4
card1
card2
D2
V96
D15
0.00
0.05
0.10
0.15
0.20
Importance Score

XGBoost Model

- Algorithm: XGBoost (Extreme Gradient Boosting) using 500 trees and a max depth of 8.
- Imbalance Handling: SMOTE Resampling - Generated 55,347 synthetic samples to balance the training set.
- Cost-Sensitive Learning: Set scale_pos_weight = 26.9 to penalize missed fraud more heavily.
- Performance Metrics:
  ROC-AUC: 0.9198 (Strong overall discriminative power).
  Recall: 72.66% (Catches nearly 3/4 of all fraud cases).
  Precision: 31.11% (Trade-off: higher false-alarm rate to ensure security).
- Top 15 Predictors: Model reliance is concentrated in specific "V" (Vesta behavioral) and "C" (Count) features, which provide significantly more information than the raw transaction amount.
- Efficiency: Entire training process completed in 35.8 seconds with total features used - 217, making it highly scalable for large datasets.



XGBoost Model Statistics

Trees: 500
Max Depth: 8
Learning Rate: 0.05
Scale Pos Weight: 26.94
Features Used: 217
Top Feature: V70

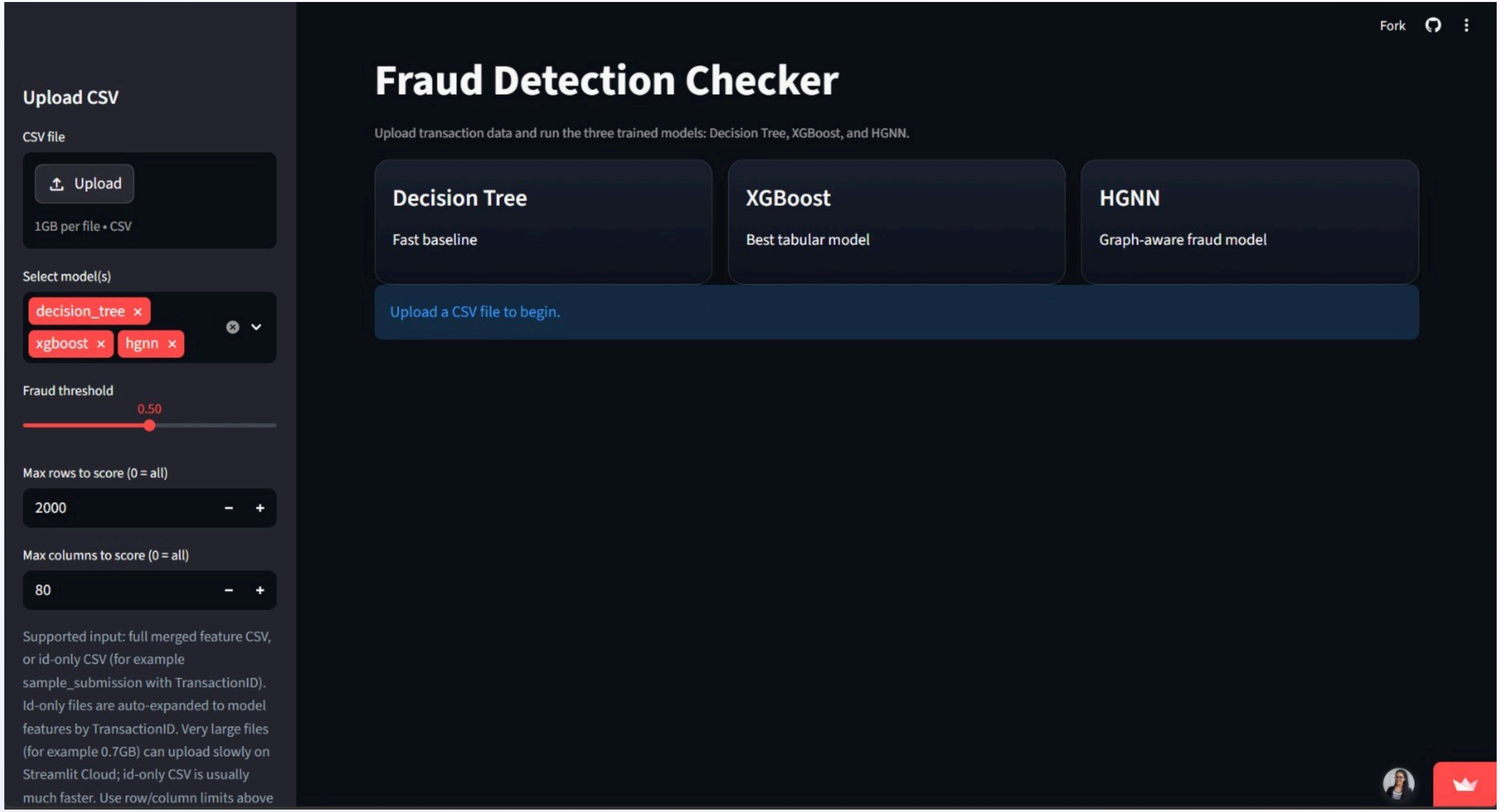
HGNN Model

- Graph-Based Fraud Architecture
- Concept: Moves beyond tabular analysis to Relationship Mapping.
- Heterogeneous Graph: Connects three distinct node types: Transactions, Cards, and Devices.
- Attention Mechanism: Uses a Transformer-based architecture to automatically weigh the importance of different connection types (e.g., identifying "High-Risk Devices").
- Temporal Intelligence: Incorporates Temporal Decay to prioritize recent behavioral patterns over historical data.
- Ideal Use Case: Best used as a high-confidence filter to automatically block transactions without needing manual review.

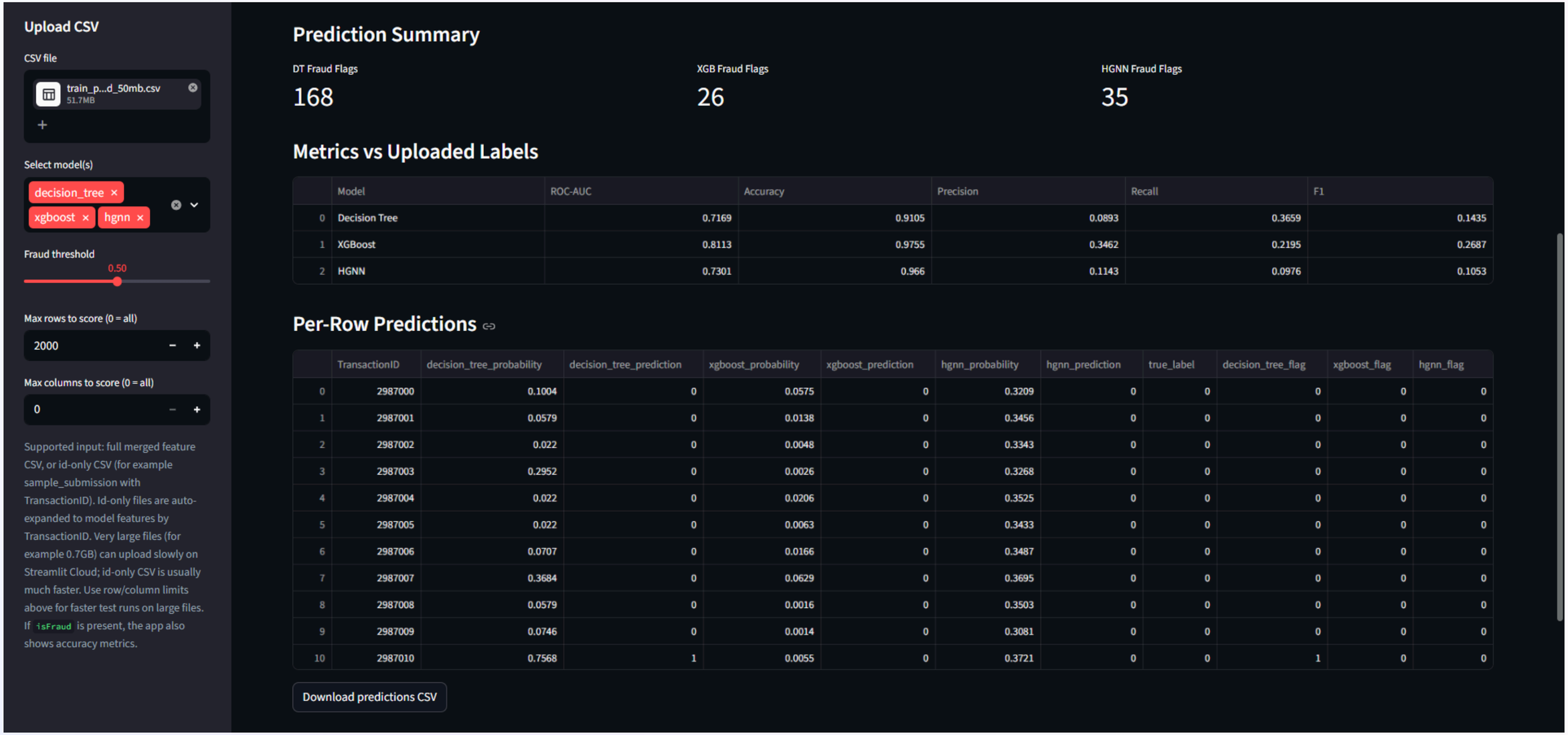
Training Curves

Preview

Deployed via Streamlit:
https://credit-card-fraud-detection-system-hyyu5xm7adgwjhhweyg8gt.streamlit.app/



Prediction Preview



Upload CSV

CSV file

train_p...d_50mb.csv
51.7MB

Select model(s)

decision_tree × xgboost × hgnn ×

Fraud threshold

0.50

Max rows to score (0 = all)

2000

Max columns to score (0 = all)

0

Supported input: full merged feature CSV, or id-only CSV (for example sample_submission with TransactionID). Id-only files are auto-expanded to model features by TransactionID. Very large files (for example 0.7GB) can upload slowly on Streamlit Cloud; id-only CSV is usually much faster. Use row/column limits above for faster test runs on large files. If `isFraud` is present, the app also shows accuracy metrics.

Prediction Summary

DT Fraud Flags
168

XGB Fraud Flags
26

HGNN Fraud Flags
35

Metrics vs Uploaded Labels

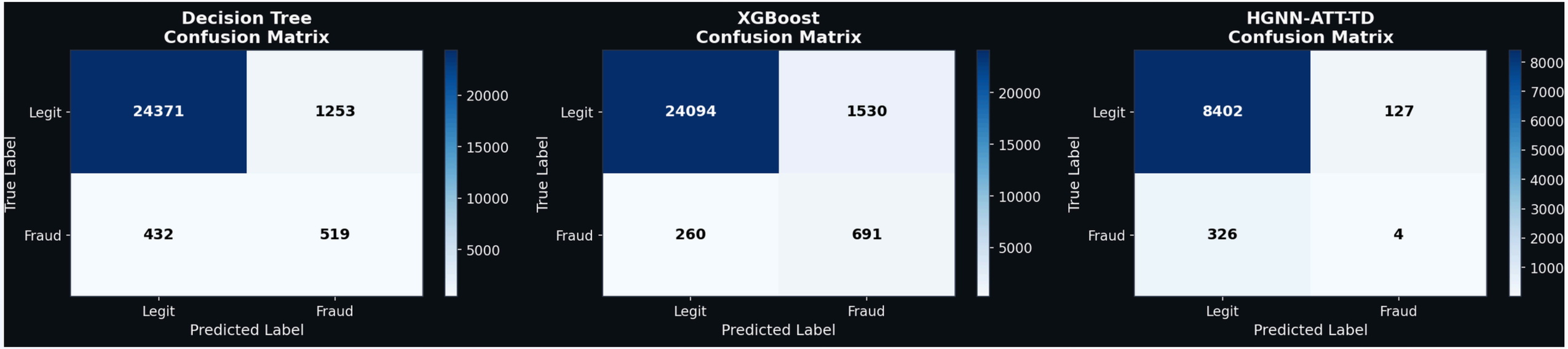
	Model	ROC-AUC	Accuracy	Precision	Recall	F1
0	Decision Tree	0.7169	0.9105	0.0893	0.3659	0.1435
1	XGBoost	0.8113	0.9755	0.3462	0.2195	0.2687
2	HGNN	0.7301	0.966	0.1143	0.0976	0.1053

Per-Row Predictions

	TransactionID	decision_tree_probability	decision_tree_prediction	xgboost_probability	xgboost_prediction	hgnn_probability	hgnn_prediction	true_label	decision_tree_flag	xgboost_flag	hgnn_flag
0	2987000	0.1004	0	0.0575	0	0.3209	0	0	0	0	0
1	2987001	0.0579	0	0.0138	0	0.3456	0	0	0	0	0
2	2987002	0.022	0	0.0048	0	0.3343	0	0	0	0	0
3	2987003	0.2952	0	0.0026	0	0.3268	0	0	0	0	0
4	2987004	0.022	0	0.0206	0	0.3525	0	0	0	0	0
5	2987005	0.022	0	0.0063	0	0.3433	0	0	0	0	0
6	2987006	0.0707	0	0.0166	0	0.3487	0	0	0	0	0
7	2987007	0.3684	0	0.0629	0	0.3695	0	0	0	0	0
8	2987008	0.0579	0	0.0016	0	0.3503	0	0	0	0	0
9	2987009	0.0746	0	0.0014	0	0.3081	0	0	0	0	0
10	2987010	0.7568	1	0.0055	0	0.3721	0	0	1	0	0

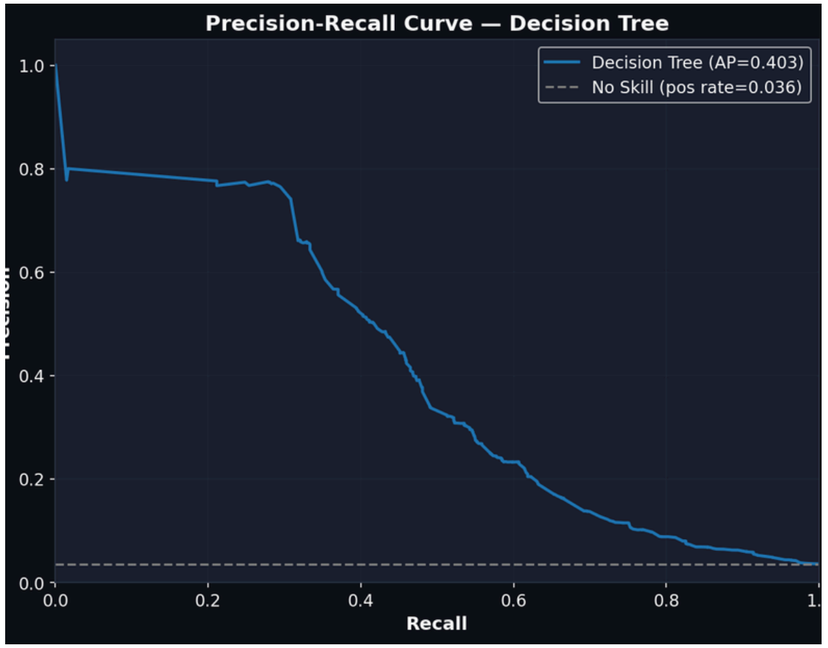
Download predictions CSV

Model Evaluation
Confusion Matrices

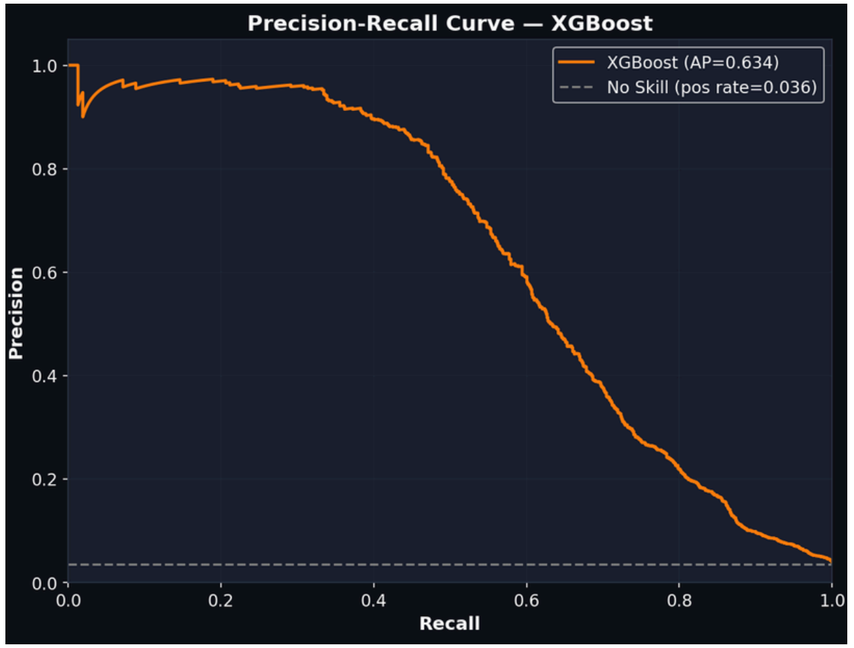


Precision Recall Curves

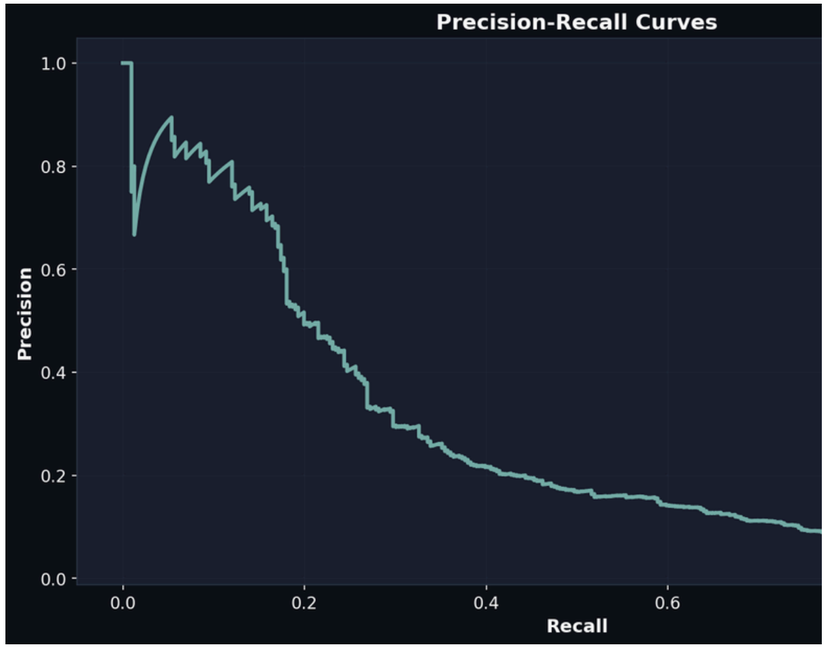
Decision Tree



XGBoost



HGNN



Metric	Decision Tree	XGBoost	HGNN-ATT-TD
Accuracy	~91%	93.26%	**96.42%**
Recall (Fraud Catch Rate)	~65%	**72.66%**	20.57%
Precision	~28%	31.11%	**49.62%**
False Positive Rate	High (~8%)	Moderate (5.97%)	**Ultra-Low (0.77%)**
Training Speed	**Fastest (< 10s)**	Fast (35.8s)	Slow (Minutes, GPU needed)

Conclusion

- HGNN-ATT-TD (Top Performer): Achieved highest accuracy (96.60%) with lowest false positives (0.77%); best for reliable, low-friction automated systems.
- Strong Detection: Correctly predicted 35/41 cases (~85%), making it the most effective overall model.
- XGBoost (Balanced Model): Highest ROC-AUC (0.8113); best at distinguishing fraud patterns and offers strong recall for deeper analysis.
- Decision Tree (High Sensitivity): Captures more fraud cases (higher recall) but with many false positives (over-flagging)
- For a production environment, the HGNN-ATT-TD is the superior choice for automated fraud prevention.
- By analyzing relationships (Graph data) rather than just independent rows (Tabular data), it provides the most stable and reliable detection with nearly zero impact on legitimate users.